The Power of Noise: Redefining Retrieval for RAG Systems

Cuconasu, Trappolini, Siciliano, Filice, Campagnano, Maarek, Tonellotto, Silvestri

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Master AI - Information Retrieval 1

Presented by Chris Hoes & Pepijn van der Klei

Problem

Limitations of LLM's:

- Knowledge is frozen at training time
- Prone to hallucinations on factual questions
- Struggle with domain-specific or rapidly changing knowledge

> Asking an LLM about the weather today?

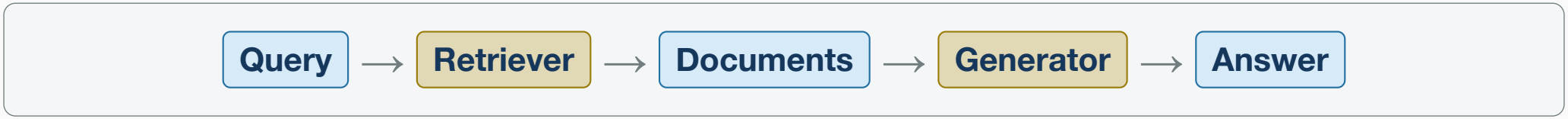
Solution?

How can we make LLMs access up-to-date knowledge?

Solution!

Retrieval-Augmented Generation (RAG):

- Use an IR system to retrieve relevant documents for the query
- Feed these documents into the LLM prompt as context



But?

Common assumption:

- Feed top-k highest-scoring retrieved docs to the LLM
- More semantically relevant == better
- Our current retrieval systems are good at this

But: the retriever's role itself has received surprisingly little systematic study.

Not So Fast

These assumptions are not actually true:

- ~~Feed top-k highest-scoring retrieved docs to the LLM~~
- ~~More semantically relevant == better~~
- ~~Our current retrieval systems are good at this~~

> We already discussed this in class, and this paper was referenced as evidence that common retrieval assumptions break down in the RAG setting.

So what **does** make a good retriever for RAG? Let's look at the study.

Research Question

> *"What characteristics are desirable in a retriever to optimize prompt construction for RAG systems? Are current retrievers ideal?"*
>
> Three dimensions studied: **type**, **position**, and **number** of documents.

Contributions:

1. First comprehensive study on how the *type* of retrieved documents affects LLM performance in RAG
2. New retrieval heuristics based on the (unexpected) results
3. All code and data publicly released

Outline

To answer this research question, the study conducts four sub-experiments:

Experiment 1: Position of the answer-containing document

Does it matter where in the prompt the answer-containing document is placed?

Experiment 2: Distracting documents

What happens when semantically close, but non-answer documents are added?

Experiment 3: Random noise

Can adding completely random documents actually help?

Experiment 4: Random noise (in a more realistic setting)

Does the noise effect hold in a realistic end-to-end RAG pipeline?

We will first cover the experimental setup, then go through each sub-experiment.

Document Taxonomy

The paper formalises four document types a retriever can return:

G (Gold) - the **ground-truth passage** containing the answer.

R (Relevant) - other passages that **also contain the answer** (not directly tested).

D (Distracting) - semantically close to the query but **no answer** (top-k but wrong).

N (Random/Noise) - **unrelated** to the query; noise.

Document Taxonomy

Prompt structure notation:

I Task instruction

———————-

G Gold document
R Relevant documents
D Distracting documents
N Noisy documents

———————-

Q The query

Example: `[I, D, G, Q]` means: instruction, then distracting docs, then gold, then query.

> The **position** of each document relative to Q will matter greatly.

Experimental Setup

Dataset

NQ-open (Natural Questions)

- 21M Wikipedia passages (100 words each)
- 2,889 test queries
- Answers ≤ 5 tokens
- Open-domain Q&A

Generators

- **Llama2-7b** (main model)
- Phi-2
- Falcon-7b
- Mosaic MPT-7b

All open-source, 4-bit quantised
Greedy decoding, max 15 tokens

Retrievers

- **Contriever** (dense, bi-encoder)
- **BM25** (sparse, term-based)

Dense vs. sparse comparison

Metric: Accuracy

Experiment 1: Gold Position

Three positions tested for the gold document **G (Gold)** relative to the query **Q**:

Near: `[I, D, G, Q]`

Gold adjacent to query

Far: `[I, G, D, Q]`

Gold at start of prompt

Mid: `[I, D, G, D, Q]`

Gold buried in the middle

Near > Far > Mid. The middle of the context is hardest to attend to.

Experiment 1: Gold Position

Intuition from attention patterns:

- The **"Lost in the Middle"** effect (Liu et al., 2023)
- Attention scores are **highest near the query**, and also relatively high at the very start of the prompt

Practical implication: place relevant docs **immediately before** the query. Start of prompt is second best.

Consistent across all tested models (Llama2, Phi-2, Falcon, MPT).

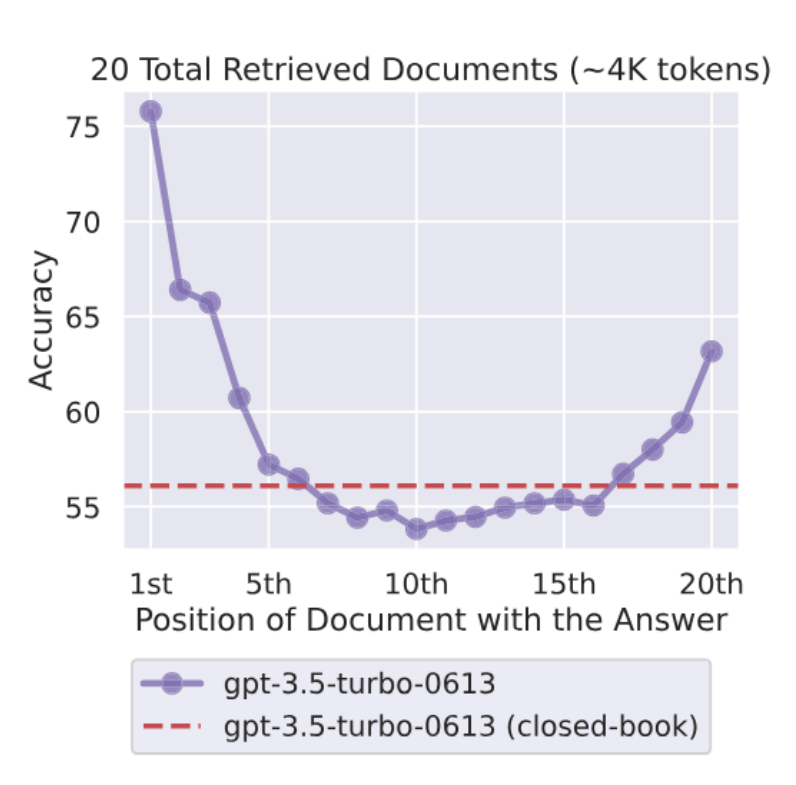


Figure 1:

Experiment 2: Distracting Documents

Common perception:

- Retrieve top-k documents by similarity score
- Semantically close documents are helpful for the LLM
- More top-scoring documents → better answers

> In practice, retrievers often return documents that are **semantically aligned** with the query but **do not contain the answer**. These are called **distracting documents** D (Distracting).

Experiment 2: Distraction Example

Query: *"Who owned the Millennium Falcon before Han Solo?"*

D (Distracting) **Doc 1:**

Before the events of the film, he and Chewbacca had lost the "Millennium Falcon" to thieves, but they reclaim the ship after it…

D (Distracting) **Doc 2:**

The "Falcon" has been depicted many times in the franchise, and ownership has changed several times…

G (Gold) **Doc 3:**

Han Solo won the Millennium Falcon from **Lando Calrissian** in the card game sabacc…

Answer: Han Solo (correct: Lando Calrissian)

Experiment 2: Distraction Setup

Sub-question:

> How do distracting documents **D (Distracting)** affect RAG performance?

Setup:

- Oracle setting: the gold document **G (Gold)** is always in the prompt
- Add 0-10 distracting documents (high retrieval score, no answer)
- Test across all four LLMs and three positions (far, mid, near)
- Schematically: `[I, D..., G, Q]`

Experiment 2: Distraction Results

Baseline comparison (Llama2, Contriever, Near):

Only gold	`[I, G, Q]`	**0.564**
Gold + 1 distracting	`[I, D, G, Q]`	↓ **0.428**
Gold + 2 distracting	`[I, DD, G, Q]`	↓↓ **0.397**

Distracting documents **D (Distracting)** (high retrieval score, **no answer**) **decrease** LLM accuracy vs. using only the gold document.

The **retriever's ranking score is not a reliable signal** for downstream LLM usefulness. High retrieval score ≠ helpful in the prompt.

Experiment 2: Distraction Results (Full)

	Far - [I, ★, ꕥ, Q]				Mid - [I, ꕥ, ★, ꕥ, Q]				Near - [I, ꕥ, ★, Q]			
# ꕥ	Llama2	MPT	Phi-2	Falcon	Llama2	MPT	Phi-2	Falcon	Llama2	MPT	Phi-2	Falcon
0	**0.5642**	0.2148	**0.4438**	**0.4330**	**0.5642**	**0.2148**	**0.4438**	**0.4330**	**0.5642**	**0.2148**	**0.4438**	**0.4330**
1	0.4586	0.1976	0.3585	0.3469	no-mid	no-mid	no-mid	no-mid	0.4283	0.1791	0.4227	0.3602
2	0.3455	0.1913	0.3430	0.3246	0.3322	0.1802	0.3375	0.2823	0.3974	0.2002	0.3975	0.3111
4	0.2745	**0.2209***	0.3019	0.2670	0.2857	0.1775	0.2885	0.2378	0.3795	0.2059*	0.3701	0.2736
6	0.2898	0.2171*	0.2943	0.2392	0.2698	0.1424	0.2625	0.2103	0.3880	0.1892	0.3623	0.2656
8	0.2643	0.2077*	0.2513	0.1878	0.2268	0.1002	0.2360	0.1745	0.3748	0.1944	0.3423	0.2424
10	0.2537	-	-	-	0.2180	-	-	-	0.3716	-	-	-
12	0.2688	-	-	-	0.2382	-	-	-	0.3991	-	-	-
14	0.2583	-	-	-	0.2280	-	-	-	0.4118	-	-	-
16	0.2413	-	-	-	0.2024	-	-	-	0.3889	-	-	-
18	0.2348	-	-	-	0.1795	-	-	-	0.3781	-	-	-

- Progressive degradation across **all LLMs**, accuracy drops up to 67%
- Even **one** distracting document causes up to 25% accuracy loss
- Consistent across all three positions (far, mid, near)

- Confirmed with **ADORE** (hard-negative retriever), same degradation
- Distinguishing relevant from distracting is a hard problem that **cannot be mitigated** by changing the retrieval method

Experiment 2: Distraction Explanation

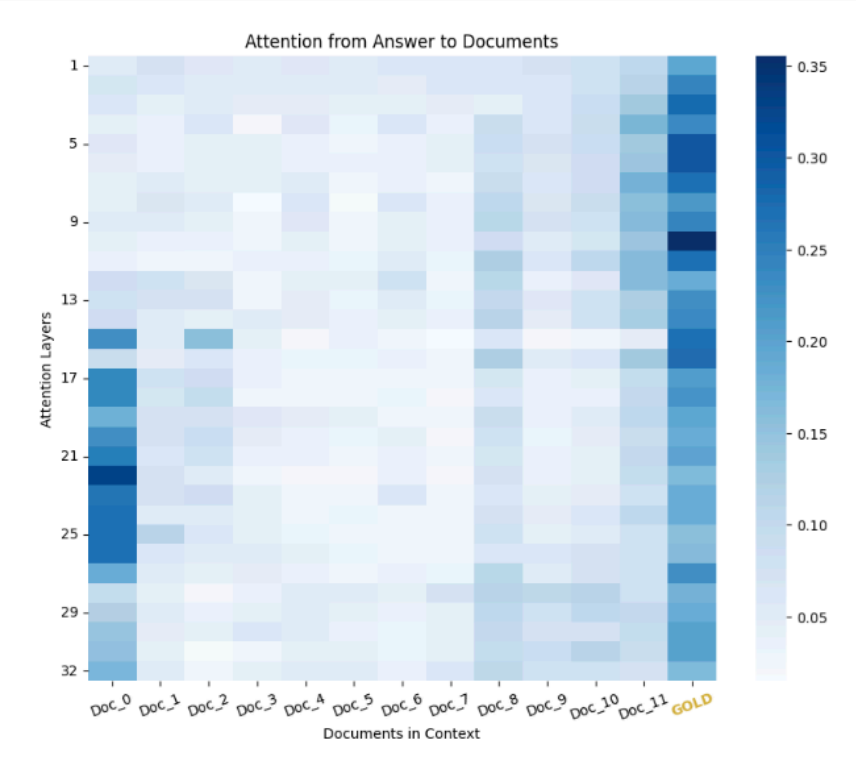


Figure 3: Attention from answer tokens to context documents in `[I, D..., G, Q]` (Llama2).

The paper's attention analysis shows the LLM **disproportionately attends** to distracting documents over the gold document, explaining the wrong answers.

> Semantically aligned documents compete for the model's attention. The LLM cannot distinguish between **relevant-with-answer** and **relevant-without-answer**.

> **Note:** the most-attended document is Doc_0 (first in prompt) and the gold is Doc_12 (last, near the query). Is this a property of distracting docs, or does position itself drive attention? LLMs are known to attend more to the start and end of the context.

Experiment 3: Noise Helps (Oracle)

The gold document is always included. Random Wikipedia passages are added around it.

Prompt: `[I, N, N, N, G, Q]`

Results (MPT-7b, Near):

Only gold `[I, G, Q]`	0.215
1. 4 random docs	**0.293**

+8% accuracy from adding documents that contain no answer at all

Experiment 3: Noise Helps (Oracle)

Not consistent across all models:

- **MPT:** noise helps in *all* positions (Near, Far, Mid)
- **Llama2 & Phi-2:** noise helps only in the **Near** setting, but performance increase is minimal
- **Falcon:** noise does *not* improve accuracy

> The benefit of noise is **model-dependent**, but Near positioning + noise is generally the best combination.

Best noise position: far from query; gold near query. `[I, N, N, N, G, Q]` outperforms `[I, G, N, N, N, Q]`.

Experiment 4: Noise in Practice

So far - oracle setting:

- Gold document G (Gold) is **always** present in the prompt
- We control exactly which documents are added
- Not representative of a real RAG pipeline

In practice:

- You retrieve top-k documents: a mix of relevant, distracting, and possibly gold
- No guarantee the gold document is even retrieved
- You don't know which document is which

> Can random noise still improve accuracy when the prompt contains **real retrieved documents** instead of a hand-picked gold document?

Experiment 4: Practice Setup

Sub-question:

> Does padding retrieved documents with random noise improve accuracy in a realistic end-to-end RAG setting?

Setup:

- Retrieve top-k documents using an IR system (no oracle, the retrieved set is a mix of **R (Relevant)**, **D (Distracting)**, and possibly **G (Gold)**)
- Pad remaining context window with random documents **N (Random/Noise)**
- Prompt: `[I, NNN, D/R/G, Q]` - noise **before** retrieved docs
- Tested on **Llama2** (Table 3) and **Falcon** (Table 5)
- Both **Contriever** (dense) and **BM25** (sparse) retrievers
- Additional noise types: **Reddit** posts and **random words** (nonsensical sentences)
- Evaluated on NQ-open **test set** (earlier experiments used training set)

Experiment 4: Practice Results

Noise still works, even better:

- Adding noise improves accuracy up to **+35%** (Contriever, 4 retrieved docs) and **+44%** (BM25)
- Almost always beneficial regardless of number of retrieved docs
- Filling the context window with noise until full yields best results

	Contriever	
without noise	retrieve 4 docs	0.187
with noise	retrieve 4 + fill rest with noise	**0.253** **(+35%)**
	BM25	
without noise	retrieve 4 docs	0.203
with noise	retrieve 4 + fill rest with noise	**0.293** **(+44%)**

Broader than expected:

- **Falcon** now benefits from noise, unlike in the oracle setting where it showed no improvement
- Holds for **BM25** too. BM25 shows 3-4 pp higher accuracy on average (higher top-k accuracy)
- **Noise source doesn't matter**: Reddit posts and random words work at least as well. Reddit even adds +9% over Wikipedia noise

Sweet spot: retrieve **3-5 docs**, pad remaining context with noise. More docs → more distractors.

Experiment 4: Detailed Results

Table 3 - Llama2: Wikipedia noise + retrieved docs (Contriever vs BM25)

	Contriever							BM25						
# 📄 / # ⋮⋮	1	2	3	4	5	8	10	1	2	3	4	5	8	10
0	0.1620	0.1866	0.1876	0.1866	0.1921	0.2198	0.2108	0.2008	0.2208	0.2084	0.2028	0.2243	0.2492	0.2447
1	0.1308	0.1616	0.1717	0.1893*	0.1987*	0.2153*	0.2146*	0.1568	0.1963	0.1921	0.2115	0.2295*	0.2475*	0.2506*
2	0.1315	0.1644	0.1859*	0.2008	0.2174	0.2156*	0.2368	0.1644	0.1973	0.2080*	0.2281	0.2558	0.2495*	0.2596
3	0.1301	0.1727	0.2008	0.2316	0.2201	0.2198	0.2409	0.1568	0.2063	0.2160	0.2520	0.2579	0.2644	0.2707
5	0.1464	0.2056	0.2233	0.2240	0.2150	**0.2451**	**0.2482**	0.1772	0.2402	0.2437	0.2520	0.2554	0.2804	**0.2866**
8	0.1734	0.2066	0.2336	0.2375	0.2454	0.2416	0.2364	0.1994	0.2451	0.2579	0.2769	0.2817	**0.2859**	0.2777
10	0.1796	0.2174	0.2450	0.2502	**0.2499**	0.2420	-	0.2108	0.2589	0.2734	0.2835	**0.2935**	0.2853	-
15	0.2018	0.2354	0.2551	**0.2530**	-	-	-	0.2243	0.2686	0.2790	**0.2928**	-	-	-
16	0.2032	**0.2471**	**0.2558**	-	-	-	-	0.2323	0.2662	**0.2838**	-	-	-	-
17	0.2039	0.2426	-	-	-	-	-	**0.2326**	**0.2693**	-	-	-	-	-
18	**0.2073**	-	-	-	-	-	-	0.2309	-	-	-	-	-	-

Rows = random docs added, columns = retrieved docs. Bold = best per column. (*) = not significant vs. baseline.

Experiment 4: Detailed Results (Falcon)

Table 5 - Falcon-7b: Reddit noise + retrieved docs `[I, NNN, D/R/G, Q]`

# 🗎 / # ⠿	1	2	3	4	5	9
0	0.1568	0.1717	0.1855	0.1938	0.1942	**0.1998**
1	0.1551*	0.1793*	0.1897*	0.1924*	0.1976*	-
2	0.1529*	0.1762*	0.1938*	0.2011*	0.1976*	-
3	0.1599*	0.1727*	0.1911*	0.2021*	**0.2118**	-
4	0.1606*	0.1758*	0.1959	0.2073	0.2108	-
5	0.1627*	0.1762*	0.2000	**0.2108**	-	-
6	0.1651*	0.1848	**0.2004**	-	-	-
7	0.1675	**0.1848**	-	-	-	-
8	**0.1682**	-	-	-	-	-

- Tested on Reddit data (Webis-TLDR-17) as noise source
- Most values marked (*), meaning improvements are **statistically significant** vs. baseline
- Best accuracy at 5 retrieved + 3 random docs: **0.2118**
- Context window limit (2048 tokens) restricts max configurations

Falcon's 2048 token context window means fewer document combinations can be tested compared to Llama2 (4096 tokens).

Experiments 3 & 4: Entropy Collapse

From prior work (Attanasio et al., 2022; Hoffmann et al., 2023):

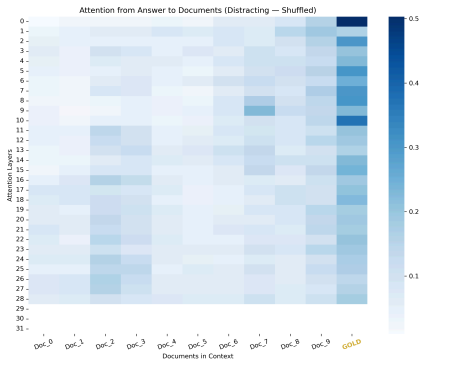
- There are cases where pathologically **low attention entropy** causes LLMs to generate degenerate outputs with a sharp decrease in performance
- These episodes are called **entropy collapse**
- Low entropy = the model focuses too narrowly on a small region of the context, ignoring relevant information

The authors' hypothesis:

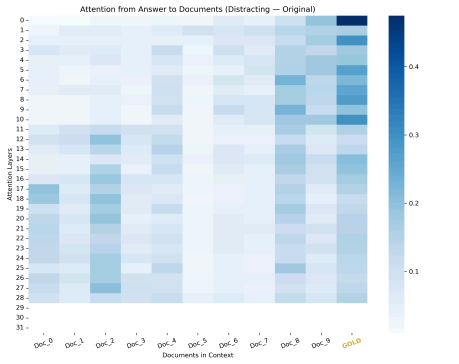
> By adding random documents to the context, we are **better conditioning** the generative function $p_\theta(y \mid \cdot, d)$, inducing enhanced accuracy. Random documents may act as "attention anchors" that **prevent entropy collapse**, spreading the model's attention more evenly and reducing degenerate focus on irrelevant passages.

Experiments 3 & 4: Entropy Evidence

Lower entropy:



Higher entropy:



Attention entropy experiment:

Measured on **Llama2-7b** in the oracle setting.

Compare the attention entropy of:

- Only gold: `[I, G, Q]`
- With noise: `[I, NN, G, Q]`

Attention entropy **increases 3x** when random documents are added to the prompt.

Higher entropy = attention is more **distributed** across documents, rather than collapsing onto a narrow, potentially misleading region.

Experiments 3 & 4: Open Question

The entropy collapse hypothesis shows **correlation** (noise → higher entropy → better accuracy) but **not causation**.

The authors explicitly note: *"although these experiments show a pattern, we cannot yet answer this question in a definitive manner."*

Experiments 1-4: Practical Heuristics

Proposed retrieval strategy:

- Retrieve **3-5 documents** with your IR system
- Place them **near the query** in the prompt
- Fill the remaining context with **random documents**, far from the query

Result: `[I, N, G/R, Q]`

Why not retrieve more?

More retrieved docs → increasing chance of **distracting** content → accuracy degrades.

Course Connection

Concepts found within paper:

Lost in the Middle

Semi-relevant documents act as a distractor

Additionally:

Entropy Collapse

> Instead of **document relevance to query**, RAG requires optimising for **LLM generation quality given the prompt**.

Conclusions

Three key findings - remember these:

- **Position matters.** Place relevant docs **near the query**. The LLM cannot attend well to distant context.
- **Distracting docs hurt.** Non-relevant top-k docs **reduce** accuracy.
- **Noise helps.** Random docs **improve** accuracy by up to 35-44%.

Code & data: `github.com/florin-git/The-Power-of-Noise`

Open questions & future work:

- **Why** does noise help? The entropy hypothesis is suggestive but not proven.
- Which properties make a document useful as filler?
- Can we **train** a retriever with an objective aligned to LLM accuracy, not just relevance?
- Do these findings hold for larger LLMs (13B, 70B)?
- Do they generalize to modern frontier models (GPT-4o, Gemini 2.5, Claude 4)?

Any Questions?

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`github.com/florin-git/The-Power-of-Noise`